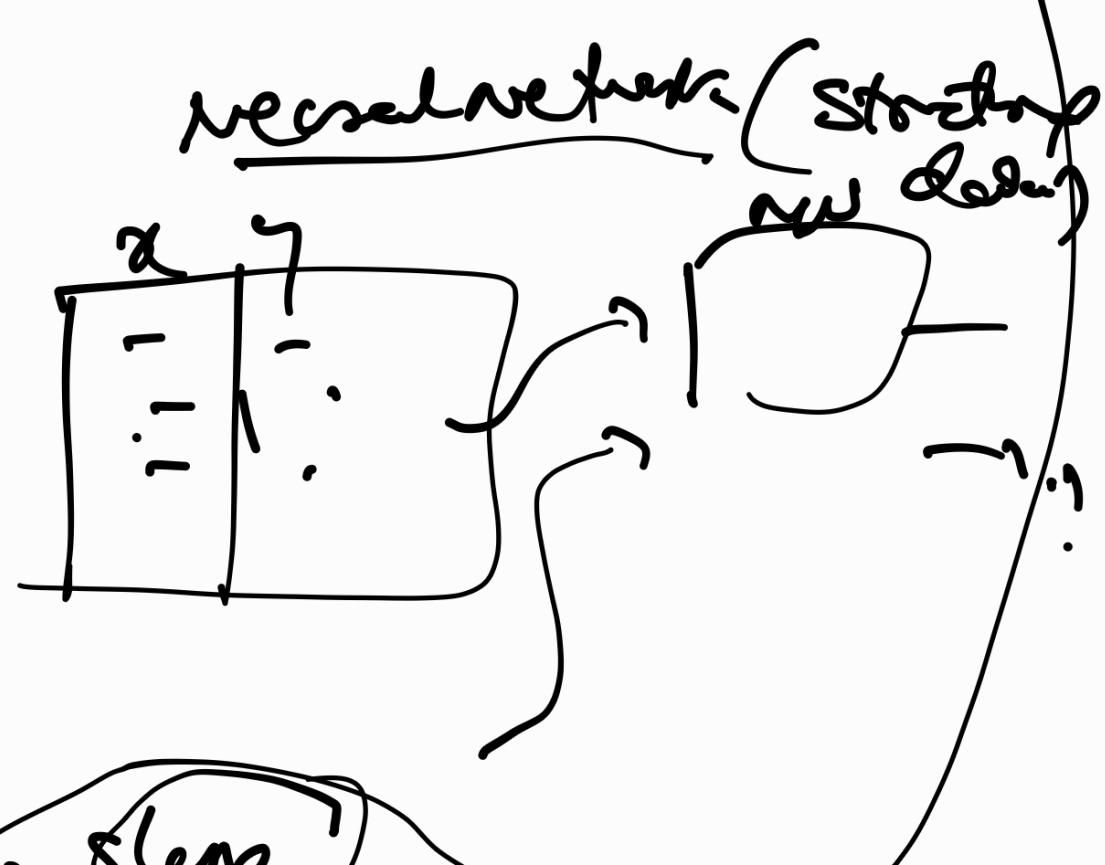
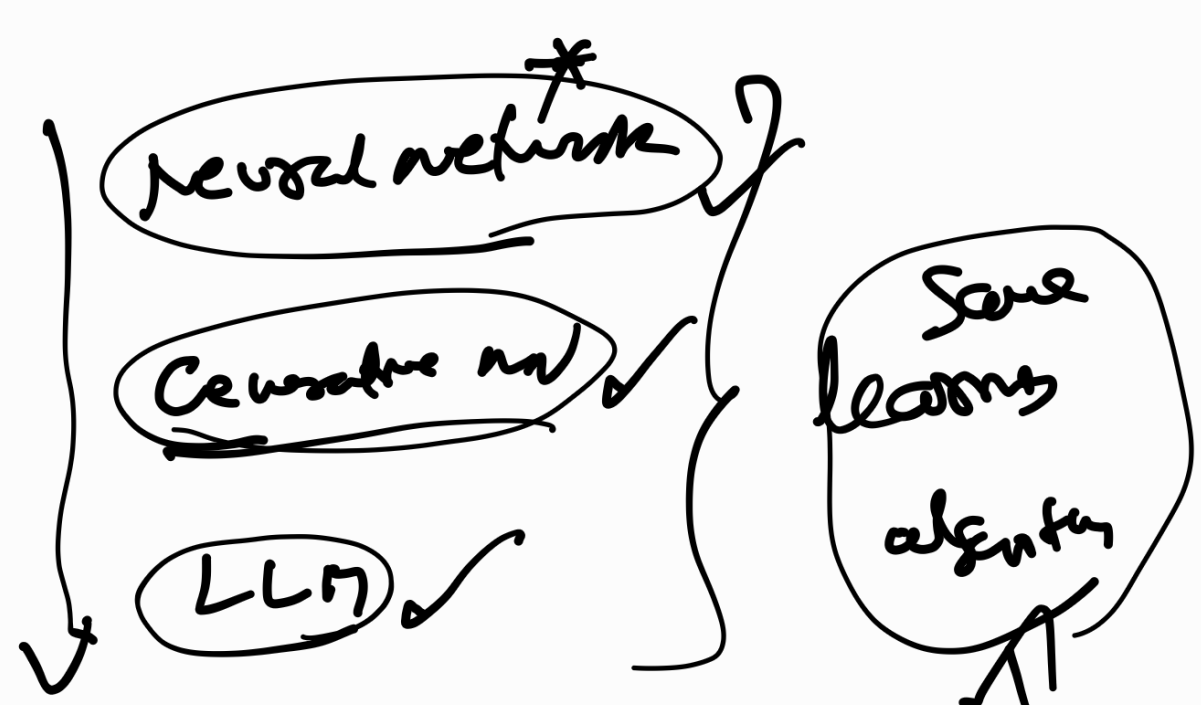
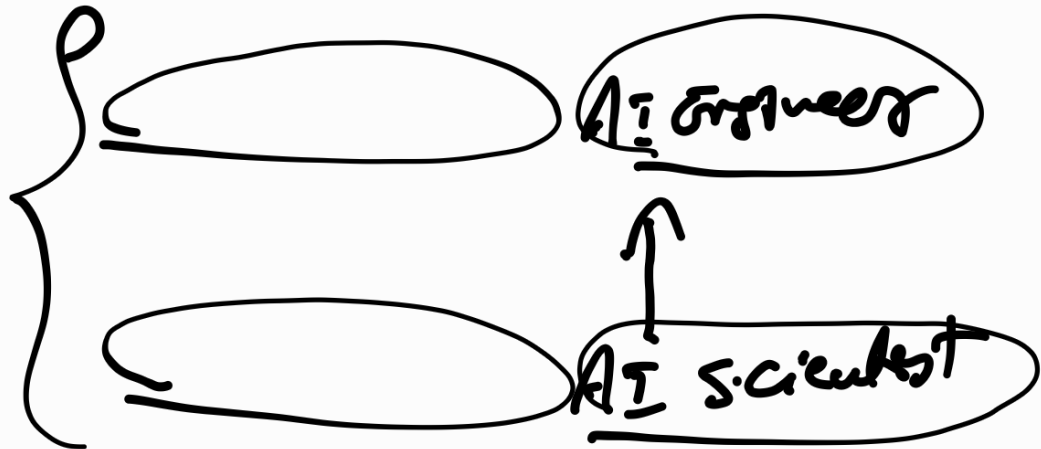




- Steps
 - Definition
 - Real world examples
 - Goal
- ↳ vector of partial derivatives
- alt

$w_0 = \dots, w_1 = \dots$

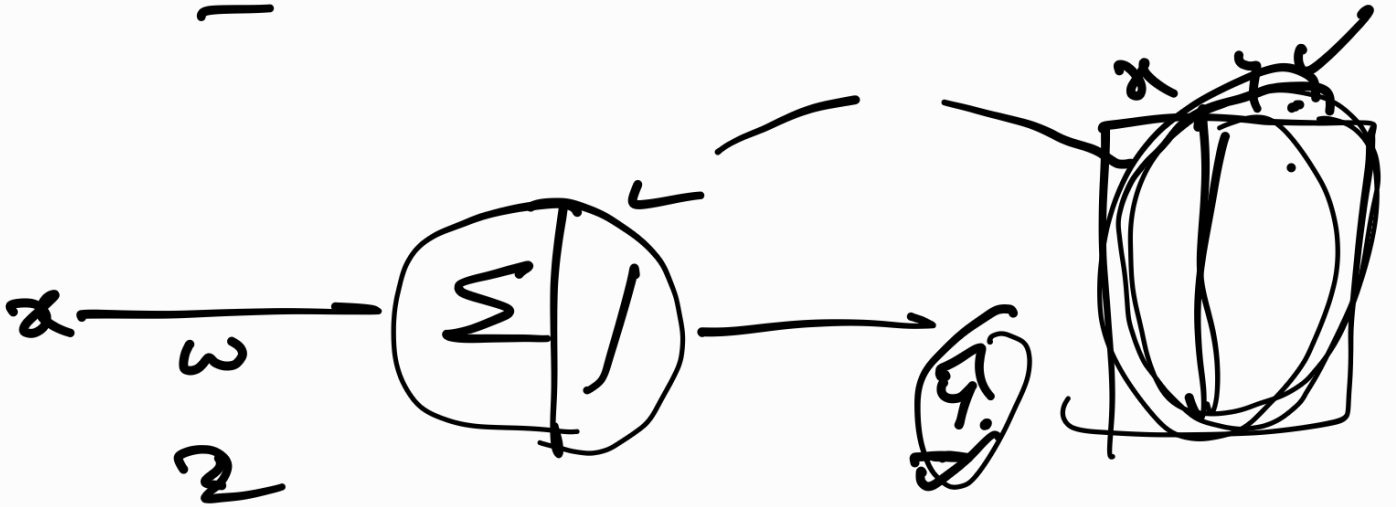
$(w_0, w_1) = (w_0, w_1) - \eta \times \left(\frac{\partial \mathcal{L}}{\partial w_0}, \frac{\partial \mathcal{L}}{\partial w_1} \right)$

A diagram showing a 3D coordinate system with axes labeled w_0 and w_1 . A point is marked on the w_1 axis, and an arrow points from it towards the origin. A checkmark is next to the w_1 axis.

batchsize
 (learns rate)
 instead of 1 by 1 weights

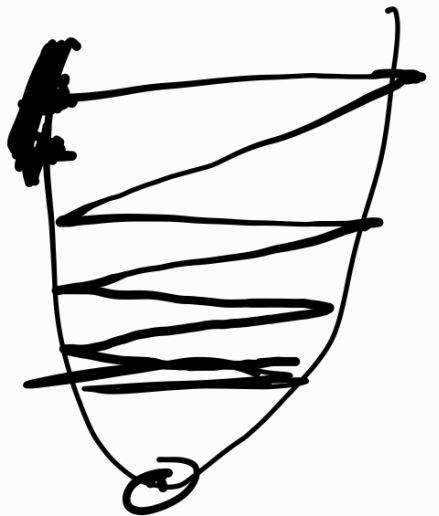
Experiment 1:

NN with 1 style neuron



$$\hat{y} = wx$$

$$w = -\hat{y}$$



$$J(w) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$\sum_{i=1}^n y_i = 10$$

let-backward

$$\frac{\partial J}{\partial w} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i) x_i$$

$$J(w) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$J(w) = \frac{1}{n} \sum_{i=1}^n (y_i - wx_i)^2$$

Issue: $\frac{1}{2000}$ is not

→ gradient/slope calculation
at each point ~~at every~~
policy whole input.

Solution

Stochastic { approximate slope calculation
using ^{random} batch of input data.

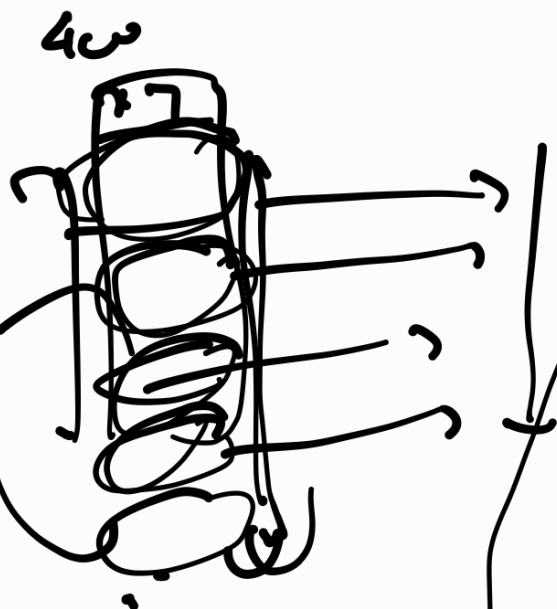
batch size = 1

→ epochs = 10000

batch size = 1000 (1/5 of 5000)

Stochastic Batch gradient descent

SGD ✓



epoch = 10000
total data

Each epoch may consist of
1 or more steps based
on batch size

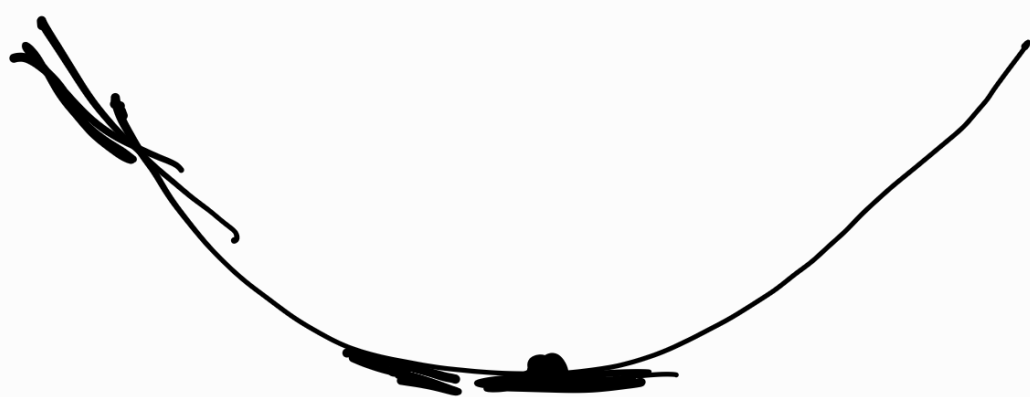
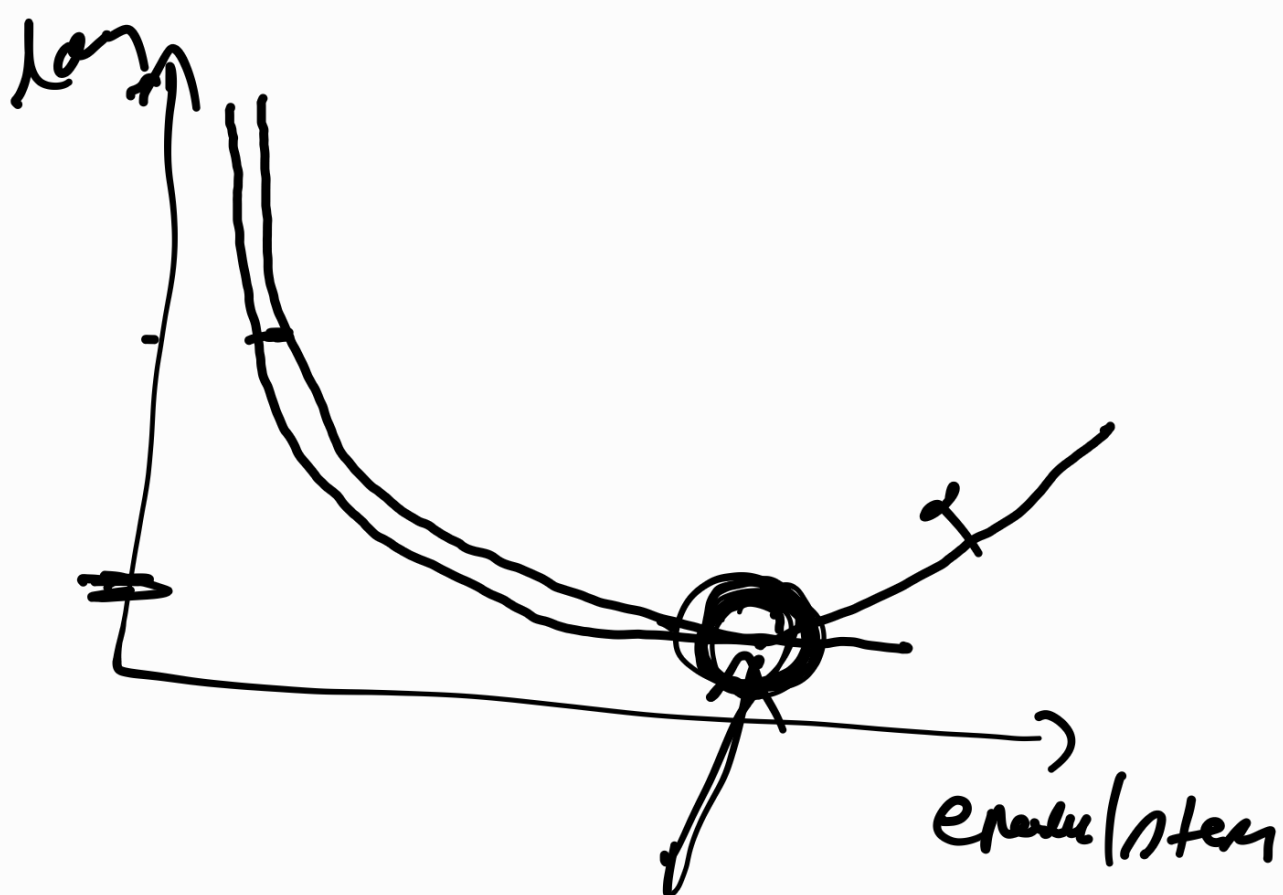
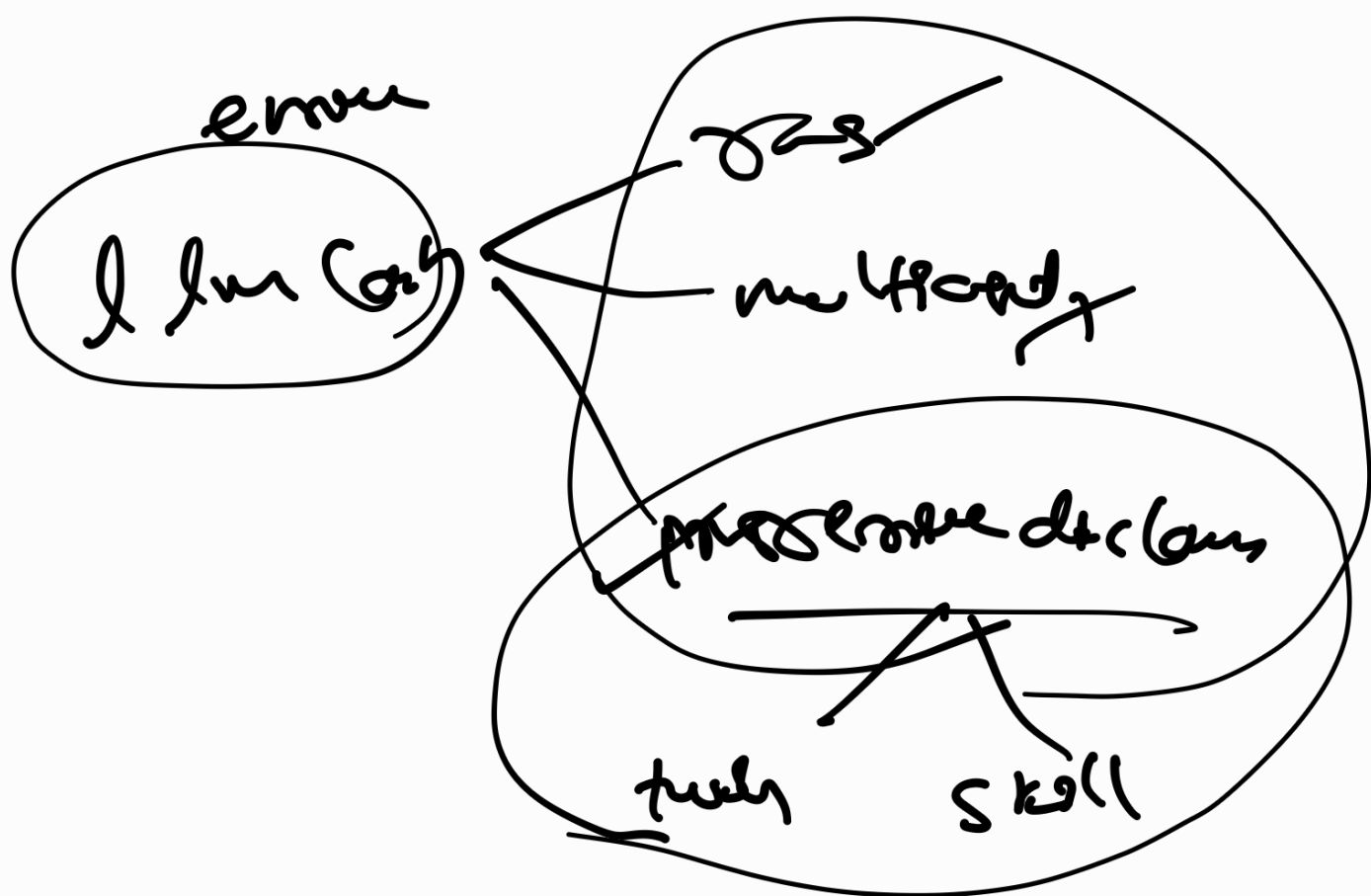
for epoch not epoch.
 for — in time: not steps
 40
 10

✓ $100 \times 40 = 4000 \text{ steps}$

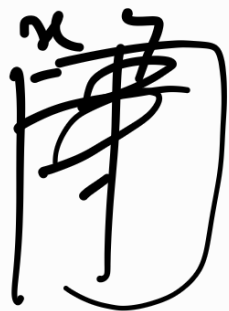
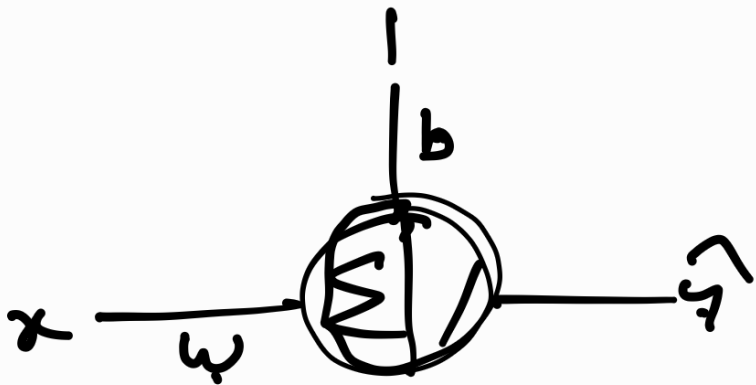
✓ $100 \times 1 = 100 \text{ steps}$

91PSA
 1,00,000
 100 epochs
 $100 \times 1 = 100 \text{ steps}$ → At each step,
 Calculate 91PSA
 1,00,000 samples.

batchsize: 32
 $100 \times \frac{32}{32} = 32,300 \text{ steps}$
 At each step,
 Calculate 91PSA
 Using 32 samples



Ex 1:



$w = -$, $b = -$

$\hookrightarrow (w, b) : (w, b) - \eta \times \left(\frac{\partial L}{\partial w}, \frac{\partial L}{\partial b} \right)$

$\swarrow$ update step()

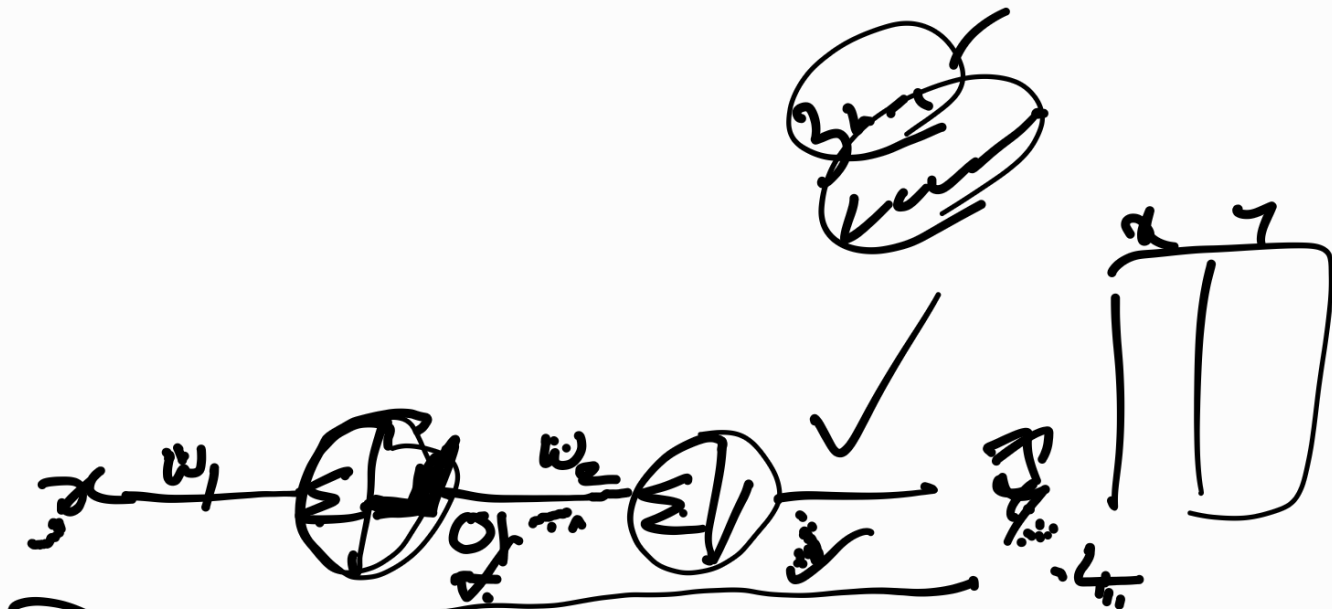
$L(w, b) = \frac{1}{n} \sum_{i=1}^n (y_i - wx_i + b)^2$

$\nwarrow$ loss function

$\frac{\partial L}{\partial w} = \frac{2}{n} \sum_{i=1}^n (y_i - wx_i + b) \cdot (-x_i)$

$\frac{\partial L}{\partial b} = \frac{2}{n} \sum_{i=1}^n (y_i - wx_i + b) \cdot 1$

Ex 2:



$L(w_1, w_2) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$

$$L(w_1, w_2) = \frac{1}{n} \sum_{i=1}^n (y_i - \underbrace{f(w_1, x_i) \cdot w_2}_{\text{forward pass}})^2$$

$$\frac{\partial L}{\partial w_1} =$$

$$\frac{\partial L}{\partial w_2} =$$

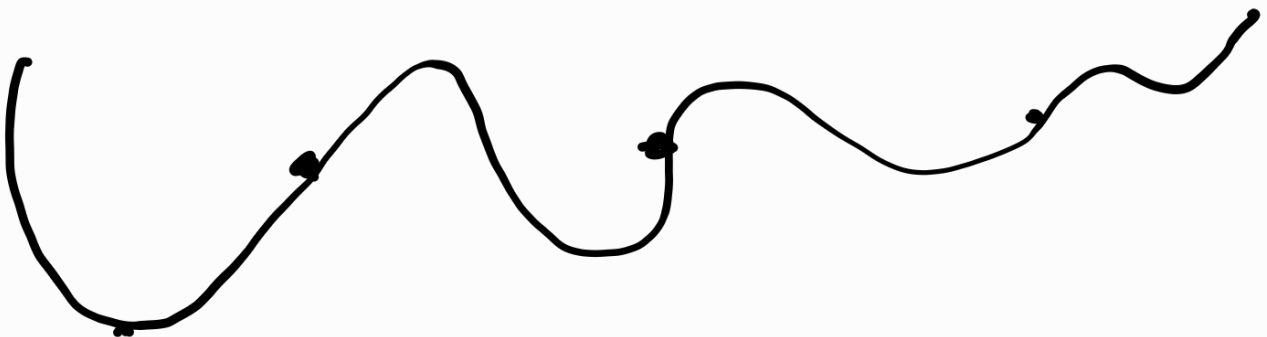
order of partial derivatives
(backward direction)

$$\sum_{i=1}^n y_i = \frac{1}{1 + \epsilon^2}$$

$$C(w_1, w_2) = (L, J(w_1, w_2)) = \mathcal{N} \times \left(\frac{\partial L}{\partial w_1}, \frac{\partial L}{\partial w_2} \right)$$

opt. step.

back. pass

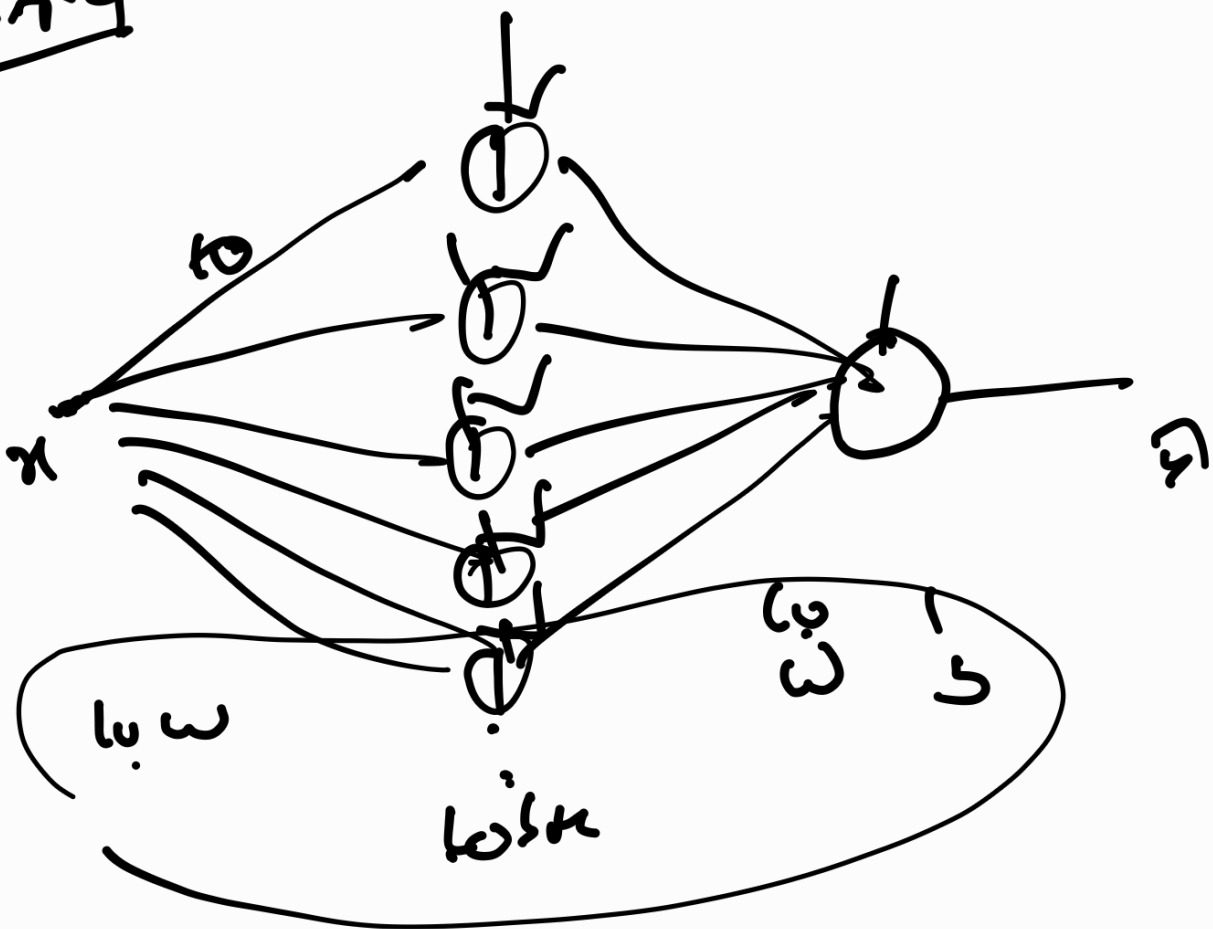


run to w

new complexity
batch size ✓
lr ✓
epoch

, normalization

eq 2



$[w] =$

$(b) =$

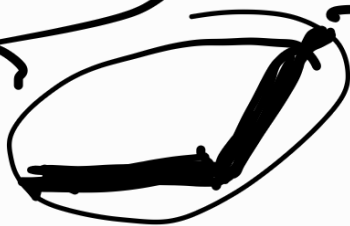
$$C(w, b) = C(w, b) - \eta \left(\frac{\partial L}{\partial w}, \dots \right)$$

f_1 forward pass
 f_2 backward pass

calculate the partial derivatives in back-propagation.

N/w size
 batch size
 epochs
 etc.

hyper parameters



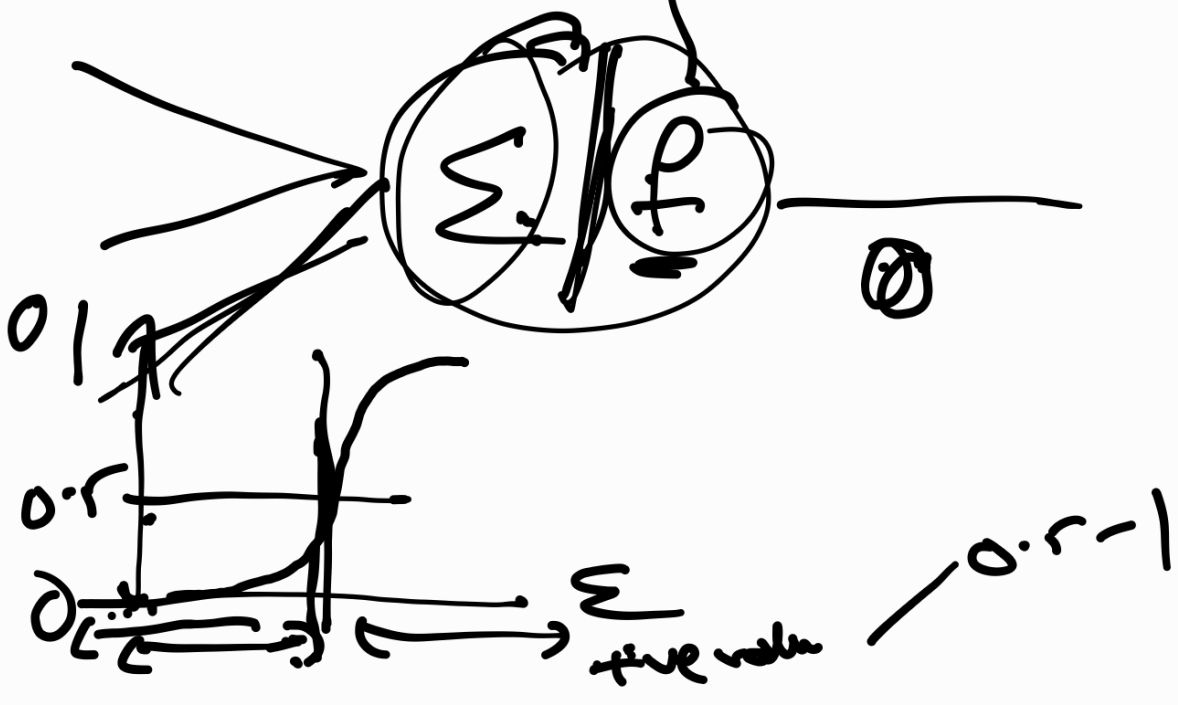
see the loss
 w
 (σw)

Sigmax

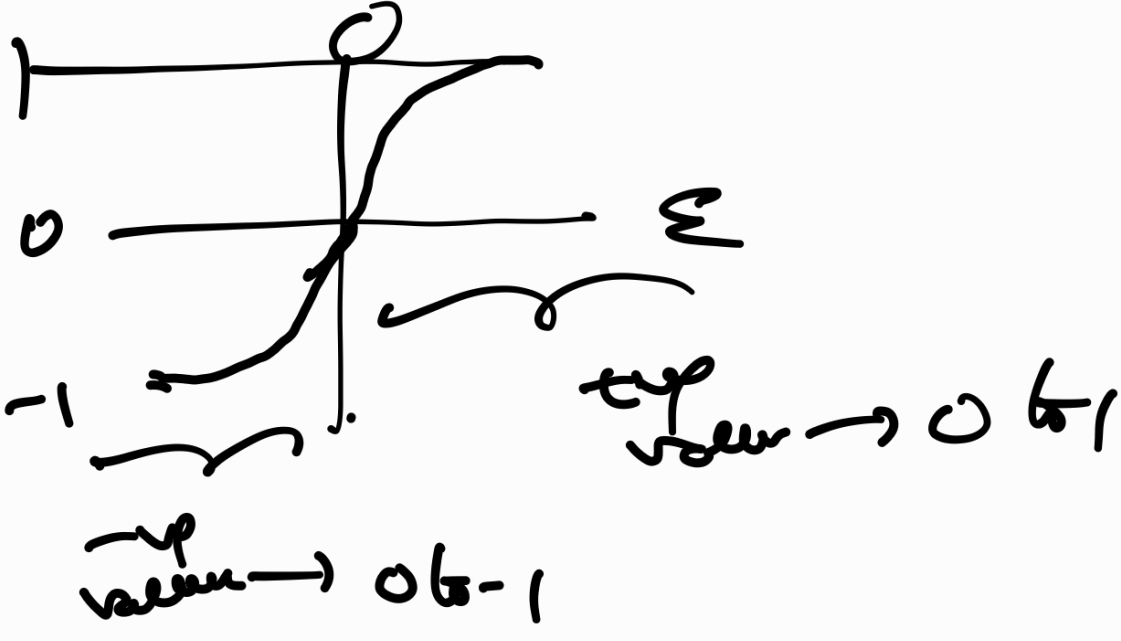
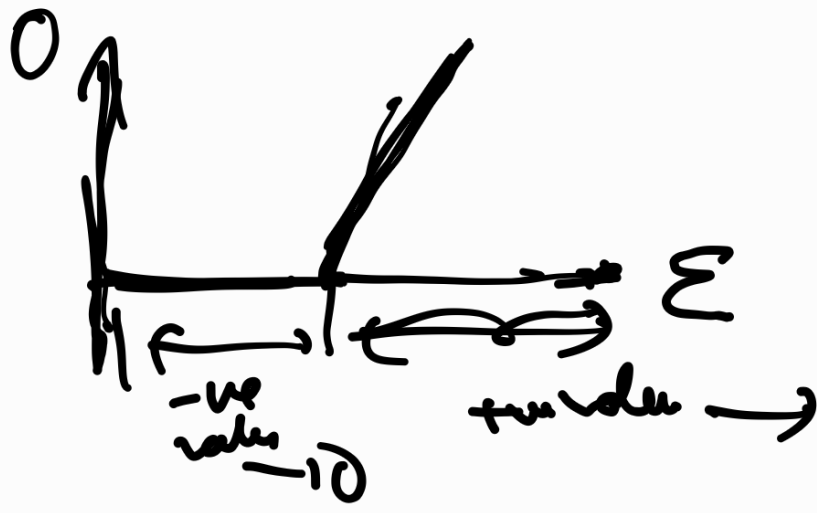
tan

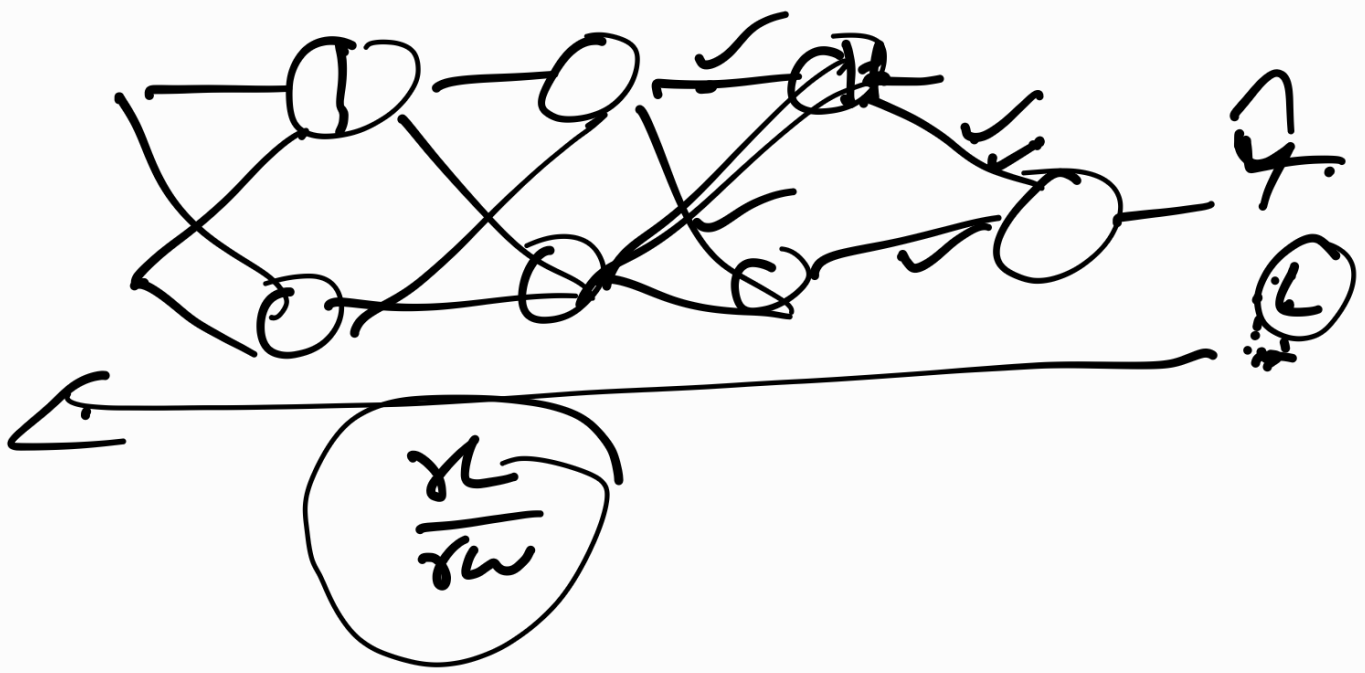
Activation
function

neuron



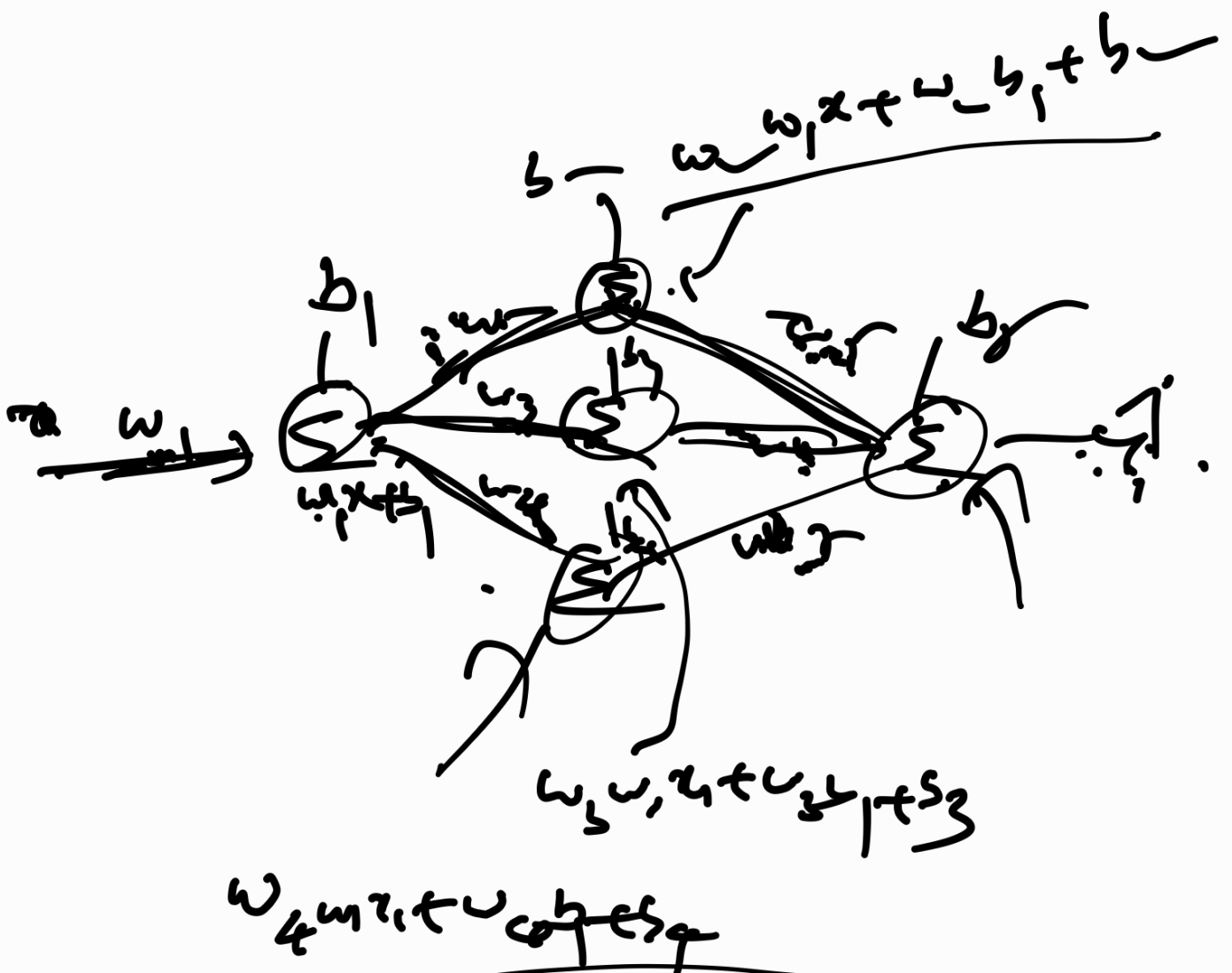
neg
values
0-0.5





fields, states, tech

- definiteness are related to Cantor
- ~~to~~ help for stable bases



$$\omega_1 \gamma - \omega_2 \gamma + \omega_1 \gamma \gamma + \omega_2 \gamma - \gamma \gamma \cdot f$$

$w \times t_5$

$$\begin{aligned}
 f_1 &= 2x + 3 \\
 f_2 &= 3x + 5 \\
 f_3 &= f_1 x + f_2 x
 \end{aligned}$$

$$\frac{\partial f_3}{\partial x} = \frac{\partial f_3}{\partial f_1} + \frac{\partial f_1}{\partial x} + \frac{\partial f_3}{\partial f_2} + \frac{\partial f_2}{\partial x}$$

$1/15 \times 2 = 10^8$
 $epoch = 10^8$

gradiente descent

10⁸ system
 each step
 new value
 der

10⁸ rule

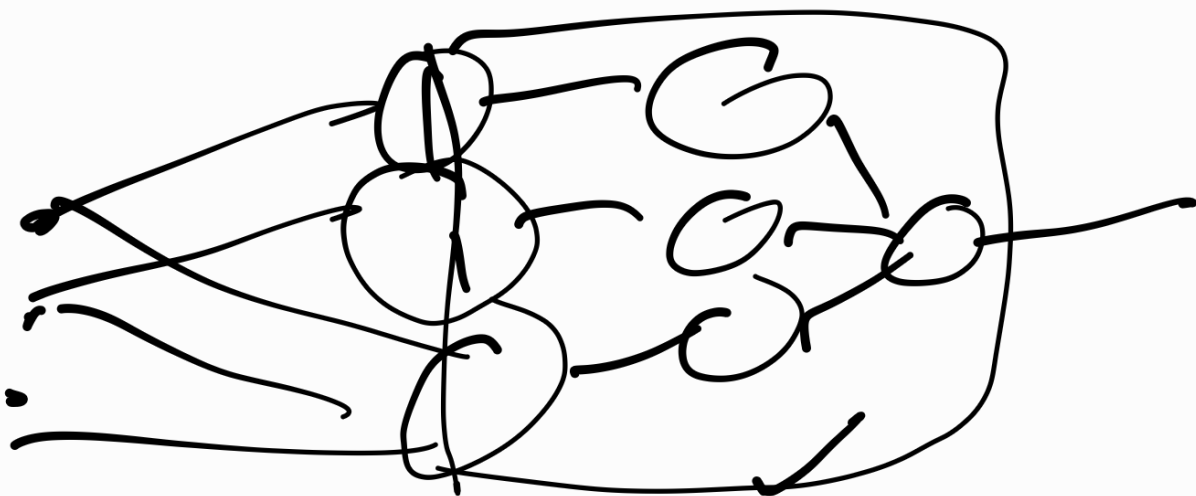
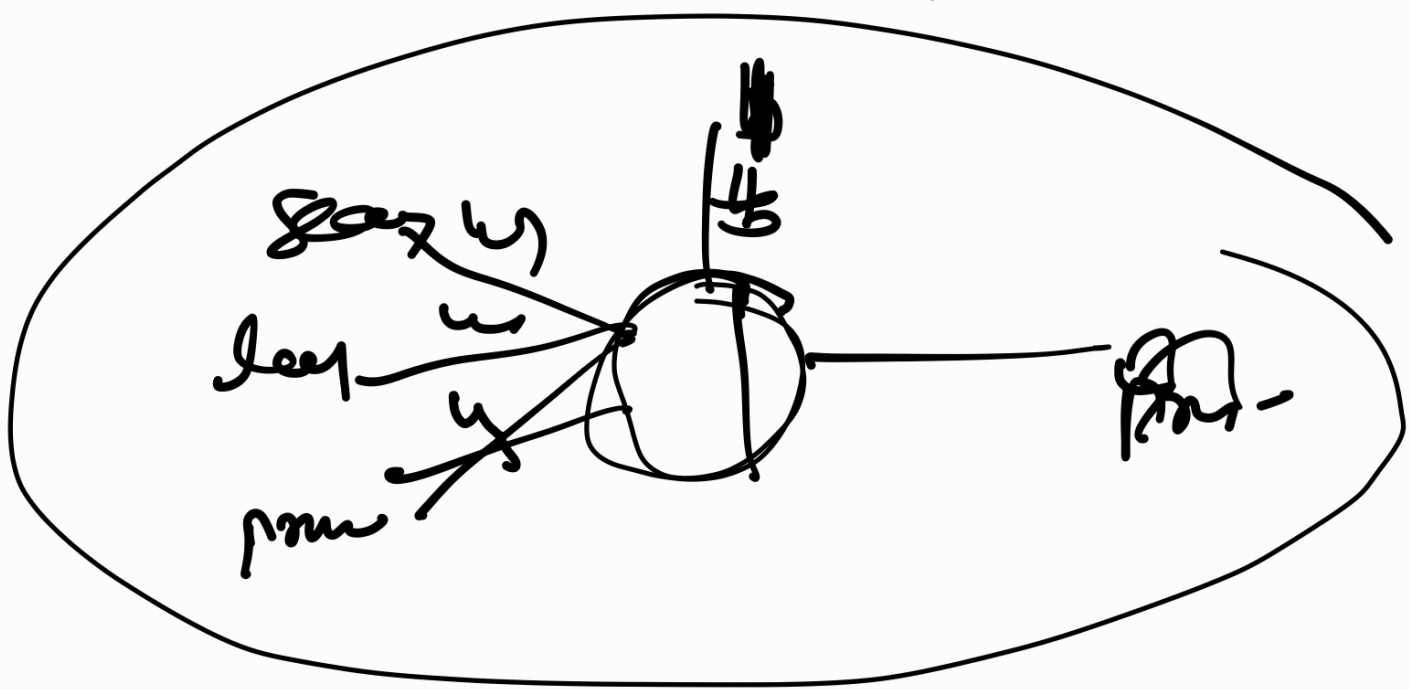
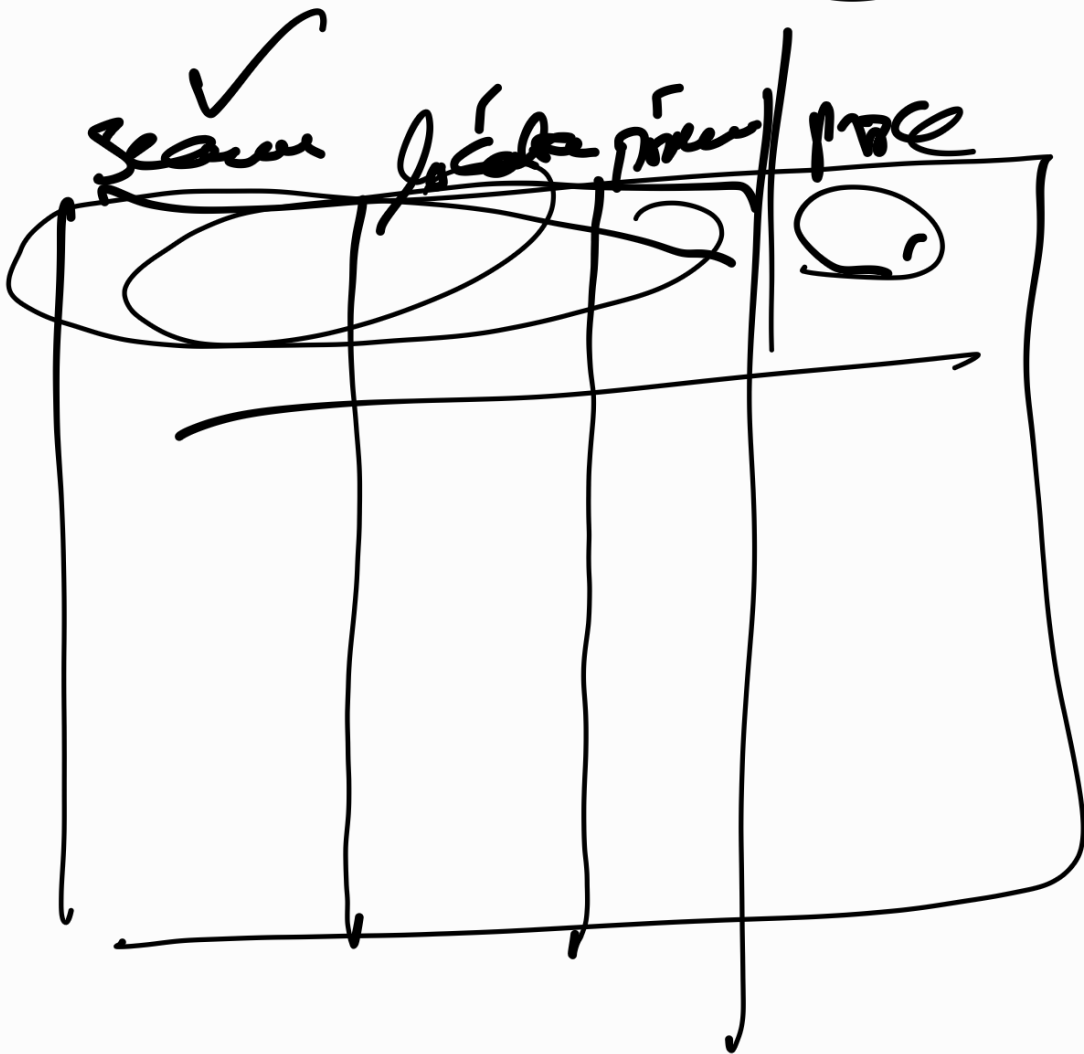
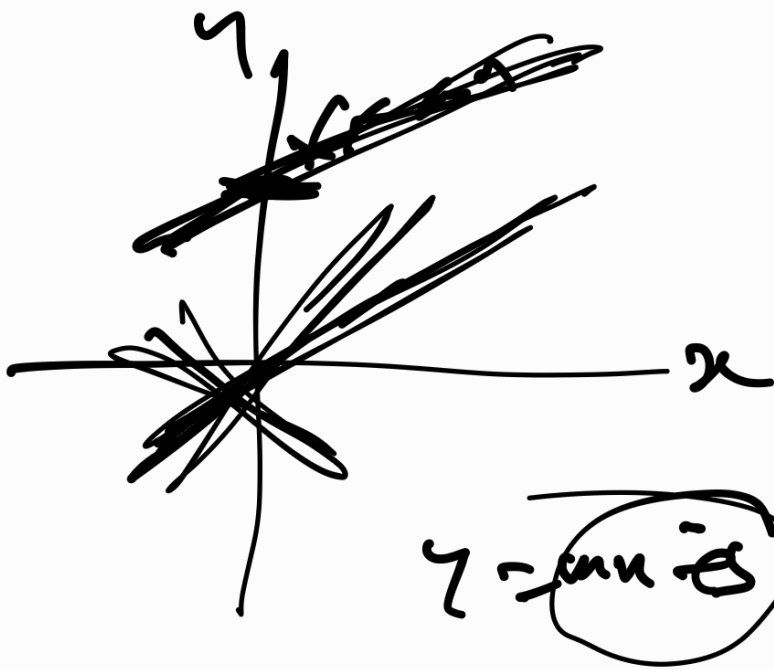
not order = $10^8 \times 10^8 = 10^{16}$

$1/15 \times 3 = 10^8$
epoch = 10⁸
 $batch \times 3 = 10^2$
Stichproben

$10^8 \times \frac{10^8}{10^8} = 10^8$

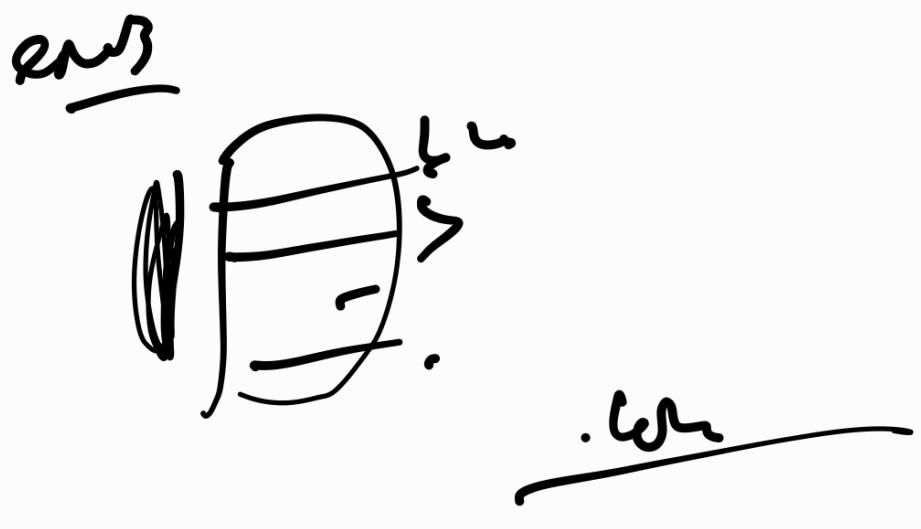
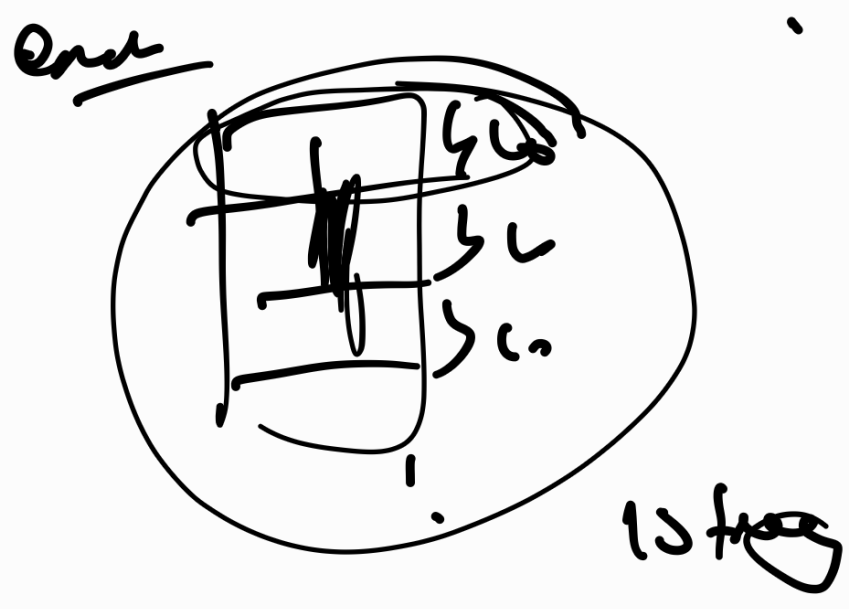
10⁸ system

$10^8 \times 10^2$ order
~~10⁸~~
10¹⁰ system





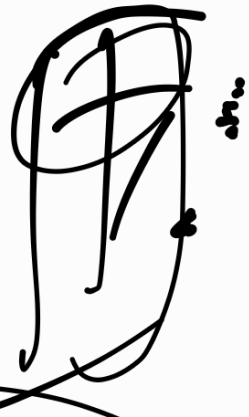
1014



$$y = \frac{1}{2}x$$

$$\frac{dy}{dx} = x$$

5. 3



$$1. (b) = \frac{1}{2} \left(\frac{1}{2} \right) \left(\frac{1}{2} \right) = \frac{1}{8}$$

$$\frac{dy}{dx} = x$$



5. 3

$$y = \frac{1}{2}x$$

$$y = \frac{1}{2}x + \frac{1}{2}$$

